Z10.5 (cont)

Corrosion is related to the ageing process, and is closely connected with the quality of corrosion prevention systems fitted at new building, and subsequent maintenance during the service life. Corrosion may also lead to cracking and/or buckling.

3.2 Methods

3.2.1 Design Details

Damage experience related to the ship in question and sister and/or similar ships, where available, is the main source of information to be used in the process of planning. In addition, a selection of structural details from the design drawings is to be included.

Typical damage experience to be considered will consist of:
- Number, extent, location and frequency of cracks
- Location of buckles.

This information may be found in the survey reports and/or the Owner's files, including the results of the Owner's own inspections. The defects are to be analyzed, noted and marked on sketches.
In addition, general experience is to be utilized. Also, reference is to be made to IACS's "Bulk Carriers: Guidelines for Survey, Assessment and Repair," (Ref. 1) which contains a catalogue of typical damages and proposed repair methods for various structural details on single skin bulk carriers. Reference should also be made to the TSCF's publication mentioned in Ref. 2, which contains catalogues of typical damages and proposed repair methods for double hull oil tanker structural details which may to some extent be similar to structural details in double skin bulk carriers. Such figures are to be used together with a review of the main drawings, in order to compare with the actual structure and search for similar details that may be susceptible to damage. In particular, Chapter 3 of Ref. 2 deals with various aspects specific to double hull tankers, such as stress concentration locations, misalignment during construction, corrosion trends, fatigue considerations and areas requiring special attention, while Chapter 4 of Ref. 2 addresses experience gained on structural defects in double hulls (chemical tankers, OBO carriers, ore/oil carriers, gas carriers), which are also to be considered in working out the survey planning.

The review of the main structural drawings, in addition to using the above-mentioned figures, is to include checking for typical design details where cracking has been experienced. The factors contributing to damage are to be carefully considered.

The use of high tensile steel (HTS) is an important factor. Details showing good service experience where ordinary, mild steel has been used may be more susceptible to damage when HTS, and its higher associated stresses, are utilized. There is extensive and, in general, good experience, with the use of HTS for longitudinal material in deck and bottom structures. Experience in other locations, where the dynamic stresses may be higher, is less favourable, e.g. side structures.

In this respect, stress calculations of typical and important components and details, in accordance with the latest Rules or other relevant methods may prove useful and are to be considered.
The selected areas of the structure identified during this process should be recorded and marked on the structural drawings to be included in the Survey Programme.

Z10.5 (cont)

3.2.2 Corrosion

In order to evaluate relative corrosion risks, the following information is generally to be considered:

- Usage of Tanks, Holds and Spaces
- Condition of Coatings
- Cleaning Procedures
- Previous Corrosion Damage
- Ballast use and time for Cargo Holds
- Risk of Corrosion in Cargo Holds and Ballast Tanks
- Location of Ballast Tanks Adjacent to Heated Fuel Oil Tanks

Ref. 3 gives definitive examples which can be used for judging and describing coating condition, using typical pictures of conditions.

The evaluation of corrosion risks is to be based on information in both Ref. 1 and Ref. 3, as far as applicable to double skin bulk carriers, together with relevant information on the anticipated condition of the ship as derived from the information collected in order to prepare the Survey Programme and the age of the ship. The various holds, tanks and spaces are to be listed with the corrosion risks nominated accordingly.

3.2.3 Locations for Close-up Survey and Thickness Measurement

On the basis of the table of corrosion risks and the evaluation of design experience, the locations for initial close-up survey and thickness measurement (areas and sections) may be nominated.

The sections subject to thickness measurement are to normally be nominated in tanks, holds and spaces where corrosion risk is judged to be the highest.

The nomination of tanks, holds and spaces for close-up survey is to, initially, be based on highest corrosion risk, and is to always include ballast tanks. The principle for the selection should that the extent is increased by age or where information is insufficient or unreliable.

Z10.5
(cont)

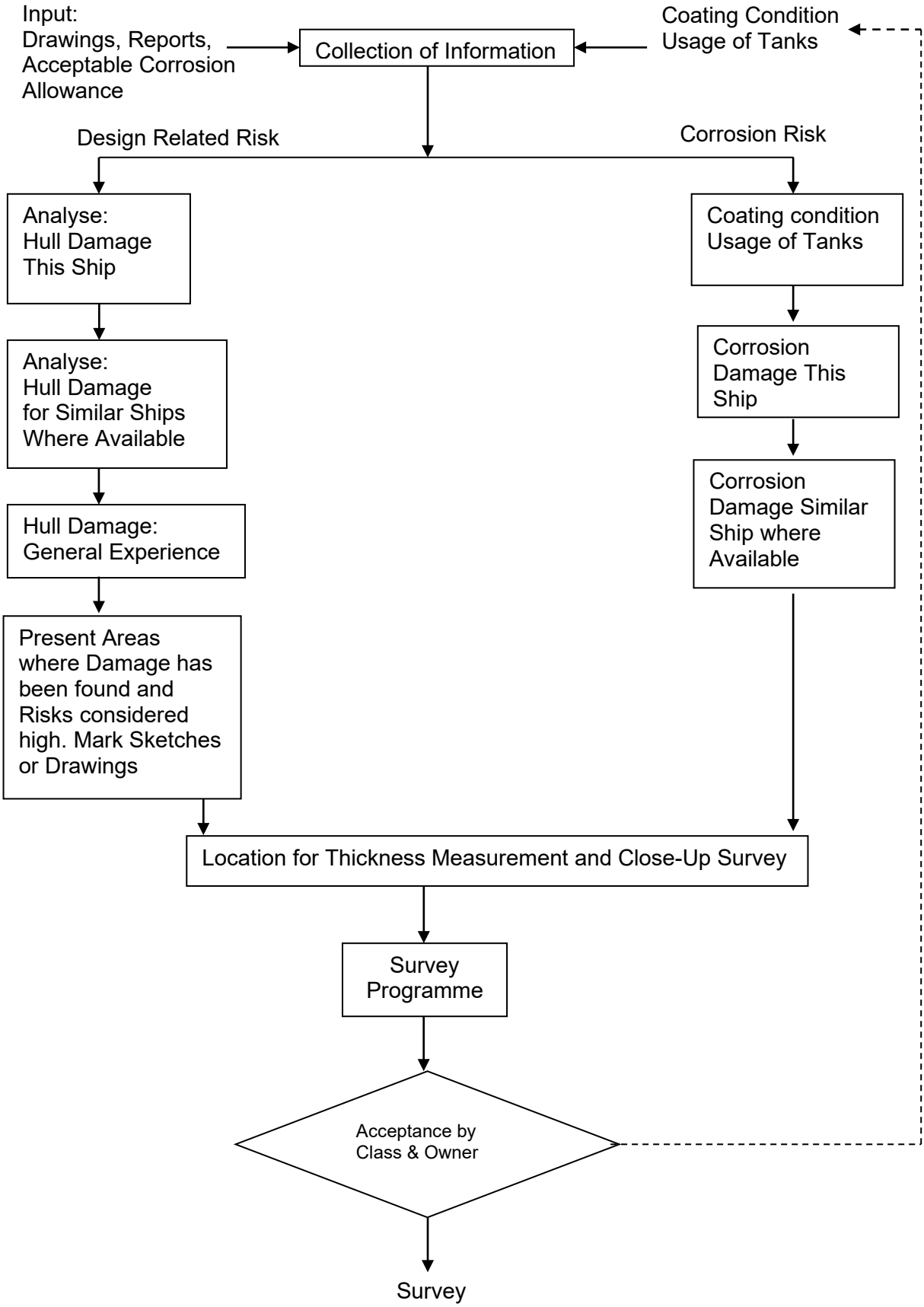

Figure 1: Technical Assessment & the Survey Planning Process

End of Annex I

Z10.5
(cont)

ANNEX II

RECOMMENDED PROCEDURES FOR THICKNESS MEASUREMENTS OF DOUBLE SKIN BULK CARRIERS *

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*
Note: Annex II is recommendatory.

Sheet 1

Z10.5
(cont)

CONTENTS

Sheet 1	-	Contents
Sheet 2	-	Instructions
Sheet 3	-	General Particulars

REPORTS

Sheet 4	-	Report TM1-DSBC for recording the thickness measurements of all deck plating, all bottom shell plating and side shell plating
Sheet 5	-	Report TM2-DSBC (i) for recording the thickness measurement of shell and deck plating at transverse sections - strength deck and sheerstrake plating
Sheet 6	-	Report TM2-DSBC (ii) for recording the thickness measurement of shell and deck plating at transverse sections - shell plating
Sheet 7	-	Report TM3-DSBC for recording the thickness measurement of longitudinal members at transverse sections (including double hull structure)
Sheet 8	-	Report TM4-DSBC for recording the thickness measurement of transverse structural members (including common frames, web frames and transverse bulkheads in double hull tanks)
Sheet 9	-	Report TM5-DSBC for recording the thickness measurement of W.T. transverse bulkheads in cargo holds
Sheet 10	-	Report TM6-DSBC for recording the thickness measurement of miscellaneous structural members

GUIDANCE

Sheet 11	-	Typical transverse section of a double skin bulk carrier. The diagram includes details of the items to be measured and the report forms to be used.
Sheet 12	-	Typical transverse section of an ore carrier. The diagram includes details of the items to be measured and the report forms to be used.
Sheet 13	-	Thickness Measurement - Double Skin Bulk Carriers Transverse section outline. The diagram may be used for those ships where the diagrams on sheet 11 and 12 are not suitable.
Sheet 14	-	Thickness Measurement - Ore Carriers Areas subject to close-up surveys and thickness measurements - areas (A) to (E) as defined in Table I of the UR Z10.5.

Z10.5
(cont)

INSTRUCTION

Sheet 2

Recommended Procedures for Thickness Measurements of Double Skin Bulk Carriers

1. This document is to be used for recording thickness measurements as required by IACS Unified Requirement Z10.5.

2. Reporting forms TM1-DSBC, TM2-DSBC, TM3-DSBC, TM4-DSBC, TM5-DSBC and TM6-DSBC (sheets 4-10) are to be used for recording thickness measurements and the maximum allowable diminution should be stated.
   The maximum allowable diminution could be stated in an attached document.

3. The remaining sheets 11-14 are guidance diagrams and notes relating to the reporting forms and the procedure for the thickness measurements.

Z10.5
(cont)

GENERAL PARTICULARS

Sheet 3

Ship's name:-

IMO Number:-

Class Identification number:-

Port of registry:-

Gross tons:-

Deadweight:-

Date of build:-

Classification Society:-

Name of Firm performing the thickness measurement:-

Thickness measurement firm certified by:-

Certificate No. :-

Certificate valid from.....to.....

Place of measurement:-

First date of measurement:-

Last date of measurement:-

Special survey/intermediate survey due:-*

Details of measurement equipment:-

Qualification of operator:-

Report Number:- Consisting of.....Sheets

Name of operator:-	.....	Name of surveyor:-	.....
Signature of operator:-.	.....	Signature of surveyor:-	.....
Firm official stamp:-	.....	Classification Society Official Stamp:-	.....

* Delete as appropriate

Z10.5 (cont)

TM1-DSBC

Report on THICKNESS MEASUREMENT of ALL DECK PLATING, ALL BOTTOM SHELL PLATING and SIDE SHELL PLATING* (* - delete as appropriate)

Sheet 4

Ship's name..... Class Identity No. ..... Report No. .....

STRAKE POSITION																	
PLATE POSITION	No. or Letter	Org. Thk. mm	Forward Reading						Aft Reading						Mean Diminution %		Maximum Allowable Diminution
			Gauged		Diminution P		Diminution S		Gauged		Diminution P		Diminution S				
			P	S	mm	%	mm	%	P	S	mm	%	mm	%	P	S	mm
12th forward																	
11th																	
10th																	
9th																	
8th																	
7th																	
6th																	
5th																	
4th																	
3rd																	
2nd																	
1st																	
Amidships																	
1st aft																	
2nd																	
3rd																	
4th																	
5th																	
6th																	
7th																	
8th																	
9th																	
10th																	
11th																	
12th																	

Operators Signature..... NOTES – See Reverse

Z10.5 (cont)

NOTES TO REPORT TM1-DSBC

1. This report is to be used for recording the thickness measurement of:-

   1.1 All strength deck plating within cargo length area.

   1.2 All keel, bottom shell plating and bilge plating within the cargo length area.

   1.3 Side shell plating including selected wind and water strakes outside cargo length area.

   1.4 All wind and water strakes within cargo length area.

2. The strake position is to be cleared indicates as follows:-

   2.1 For strength deck indicate the number of the strake of plating inboard from the stringer plate.

   2.2 For bottom plating indicate the number of the strake of plating outboard from the keel plate.

   2.3 For side shell plating give number of the strake of plating sheerstrake and letter as shown on shell expansion.

3. Only the deck plating strakes outside line of openings are to be recorded.

4. Measurements are to be taken at the forward and aft areas of all plates and where plates cross ballast/cargo tank boundaries separate measurements for the area of plating in way of each type of tank are to be recorded.

5. The single measurements recorded are to represent the average of multiple measurements.

6. The maximum allowable diminution could be stated in an attached document.

Z10.5 (cont) **TM2-DSBC (i)**

Report on THICKNESS MEASUREMENT OF SHELL AND DECK PLATING at transverse sections (one, two or three transverse sections)

Sheet 5

Ship's name..... Class Identity No. ..... Report No. .....

STRENGTH DECK AND SHEERSTRAKE PLATING																												
	FIRST TRANSVERSE SECTION AT FRAME NUMBER									SECOND TRANSVERSE SECTION AT FRAME NUMBER									THIRD TRANSVERSE SECTION AT FRAME NUMBER								
STRAKE POSITION	No. or Letter	Org. Thk.	Max. Alwb. Dim.	Gauged		Diminution P		Diminution S		No. or Letter	Org. Thk.	Max. Alwb. Dim.	Gauged		Diminution P		Diminution S		No. or Letter	Org. Thk.	Max. Alwb. Dim.	Gauged		Diminution P		Diminution S	
		mm	mm	P	S	mm	%	mm	%		mm	mm	P	S	mm	%	mm	%		mm	mm	P	S	mm	%	mm	%
Stringer Plate																												
1st strake inboard																												
2nd																												
3rd																												
4th																												
5th																												
6th																												
7th																												
8th																												
9th																												
10th																												
11th																												
12th																												
13th																												
14th																												
centre strake																												
sheer strake																												
TOPSIDE TOTAL																												

Operators Signature..... NOTES – See Reverse

Z10.5 (cont)

NOTES TO REPORT TM2-DSBC (i)

1. This report is to be used for recording the thickness measurement of:-

   Strength deck plating and sheerstrake plating transverse sections:-

   One, two or three sections within the cargo length area, comprising the structural items (0), (1) and (2) as shown on the diagrams of typical transverse sections illustrated on sheets 11, 12 and 13 of this document.

2. Only the deck plating strakes outside line of hatch openings are to be recorded.

3. The top side area comprises deck plating, stringer plate and sheerstrake (including rounded gunwales).

4. The exact frame station of measurement is to be stated.

5. The single measurements recorded are to represent the average of multiple measurements.

6. The maximum allowable diminution could be stated in an attached document.

Z10.5 (cont) **TM2-DSBC (ii)**

Report on THICKNESS MEASUREMENT OF SHELL AND DECK PLATING at transverse sections (one, two or three transverse sections)

Sheet 6

Ship's name..... Class Identity No. ..... Report No. .....

SHELL PLATING																												
	FIRST TRANSVERSE SECTION AT FRAME NUMBER									SECOND TRANSVERSE SECTION AT FRAME NUMBER									THIRD TRANSVERSE SECTION AT FRAME NUMBER								
STRAKE POSITION	No. or Letter	Org. Thk.	Max. Alwb. Dim.	Gauged		Diminution P		Diminution S		No. or Letter	Org. Thk.	Max. Alwb. Dim.	Gauged		Diminution P		Diminution S		No. or Letter	Org. Thk.	Max. Alwb. Dim.	Gauged		Diminution P		Diminution S	
		mm	mm	P	S	mm	%	mm	%		mm	mm	P	S	mm	%	mm	%		mm	mm	P	S	mm	%	mm	%
1st below sheer strake																												
2nd																												
3rd																												
4th																												
5th																												
6th																												
7th																												
8th																												
9th																												
10th																												
11th																												
12th																												
13th																												
14th																												
15th																												
16th																												
17th																												
18th																												
19th																												
20th																												
keel strake																												
BOTTOM TOTAL																												

Operators Signature..... NOTES – See Reverse

Z10.5 (cont)

NOTES TO REPORT TM2-DSBC (ii)

1. This report is to be used for recording the thickness measurement of:-

   Shell plating at transverse sections:

   One, two or three sections within the cargo length area, comprising the structural items (3), (4), (5) and (6) as shown on the diagrams of typical transverse sections illustrated on sheets 11, 12 and 13 of this document.

2. The bottom area comprises keel, bottom and bilge plating.

3. The exact frame station of measurement is to be stated.

4. The single measurements recorded are to represent the average of multiple measurements.

5. The maximum allowable diminution could be stated in an attached document.

Z10.5 (cont) **TM3-DSBC**

Report on THICKNESS MEASUREMENT OF LONGITUDINAL MEMBERS at transverse sections (one, two or three transverse sections)

Sheet 7

Ship's name..... Class Identity No. ..... Report No. .....

	FIRST TRANSVERSE SECTION AT FRAME NUMBER									SECOND TRANSVERSE SECTION AT FRAME NUMBER									THIRD TRANSVERSE SECTION AT FRAME NUMBER								
STRUCTURAL MEMBER	Item No.	Org. Thk.	Max. Alwb. Dim.	Gauged		Diminution P		Diminution S		Item No.	Org. Thk.	Max. Alwb. Dim.	Gauged		Diminution P		Diminution S		Item No.	Org. Thk.	Max. Alwb. Dim.	Gauged		Diminution P		Diminution S	
		mm	mm	P	S	mm	%	mm	%		mm	mm	P	S	mm	%	mm	%		mm	mm	P	S	mm	%	mm	%

Operators Signature..... NOTES – See Reverse

Z10.5 (cont)

NOTES TO REPORT TM3-DSBC

1. This report is to be used for recording the thickness measurement of:-

   Longitudinal Members at transverse sections:-

   Two, or three sections within the cargo length area comprising the appropriate structural items (10) to (25) as shown on diagrams of typical transverse sections illustrated on sheets 11, 12 and 13 of this document.

2. The exact frame station of measurement is to be stated.

3. The single measurements recorded are to represent the average of multiple measurements.

4. The maximum allowable diminution could be stated in an attached document.

Z10.5 (cont) **TM4-DSBC**

Report on THICKNESS MEASUREMENT OF TRANSVERSE STRUCTURAL MEMBERS
In the double bottom, hopper side and topside water ballast tanks

Sheet 8

Ship's name..... Class Identity No. ..... Report No. .....

TANK DESCRIPTION:

LOCATION OF STRUCTURE:

STRUCTURAL MEMBER	ITEM	Original Thickness mm	Max. Alwb. Dim. mm	Gauged		Diminution P		Diminution S	
				P	S	mm	%	mm	%

Operators Signature..... NOTES – See Reverse

Z10.5 (cont)

NOTES TO REPORT TM4-DSBC

1. This report is to be used for recording the thickness measurement of:

   Transverse structural members, comprising the appropriate structural items (30) to (34) as shown on diagrams of typical transverse sections illustrated on sheets 11, 12 and 13 of this document.

2. Guidance for areas of measurements is indicated on sheet 14 of this document.

3. The single measurements recorded are to represent the average of multiple measurements.

4. The maximum allowable diminution could be stated in an attached document.